

— DATA ANALYST PORTFOLIO PROJECT

# Facebook Ads vs Google AdWords

A statistical A/B test built to answer one real question: where should the next marketing dollar go?

PYTHON

PANDAS

SCIPY

HYPOTHESIS TESTING

BUSINESS ANALYTICS

## — EXECUTIVE SUMMARY

# A budget question, answered with evidence

1,000 paired campaign-day records, compared on reach, engagement, and cost-efficiency — with every observed gap validated by formal hypothesis testing, not raw averages.

**3x**

Facebook's conversion rate vs AdWords

**37%**

Lower cost-per-conversion on Facebook

**5 / 5**

Metrics statistically significant ( $p < 0.05$ )

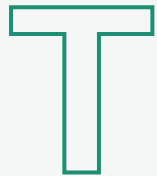
## — PROJECT NARRATIVE — STAR METHOD

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### SITUATION

A company runs the same campaigns on two platforms — Facebook and Google AdWords — under a fixed, limited marketing budget. Leadership had no reliable way to tell which platform was actually delivering better returns; decisions were made on raw averages and channel familiarity, not evidence.

## — PROJECT NARRATIVE — STAR METHOD



### TASK

Analyze 1,000 daily campaign records across both platforms, design metrics that reflect true return on spend, and determine whether the performance gap between platforms was statistically meaningful — then turn that into a budget recommendation leadership could act on.

## — PROJECT NARRATIVE — STAR METHOD



### ACTION

- Cleaned and validated the dataset — type casting, null/duplicate checks, date parsing
- Engineered a **cost-per-conversion** metric to unify performance and spend into one number
- Bucketed conversions on a shared scale so platforms were judged consistently
- Tested normality and variance *before* choosing between a t-test or Mann-Whitney U — rather than defaulting to one method

## — PROJECT NARRATIVE — STAR METHOD

### RESULT

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Facebook converts at ~3x the rate of AdWords, at a 37% lower cost per conversion. AdWords delivers ~2x the reach at a lower cost-per-click. All five metrics tested were statistically significant — turning an opinion-based debate into a defensible, evidence-backed recommendation.

— BUSINESS PROBLEM

## **Every dollar spent on one platform is a dollar not spent on the other**

Without statistical validation, teams risk shifting budget toward metrics that look different but aren't reliably different — or missing real differences hidden in daily noise.

**"Does platform choice make a measurable, statistically defensible difference — and if so, which platform earns more budget for which objective?"**

## Five steps, each one gating the next

### Dataset

1,000 daily records (2021–2023), Facebook & AdWords tracked in parallel — views, clicks, conversions, CTR, CPC, cost-per-ad. No missing values or duplicates.

1. **Clean & validate** — types, nulls, dates
2. **Explore** — distributions, correlation matrix, trends
3. **Engineer metrics** — cost-per-conversion
4. **Bucket fairly** — shared conversion bands
5. **Test rigorously** — normality → variance → correct test



— KEY FINDINGS

## The scoreboard

### ● Facebook

|                   |         |
|-------------------|---------|
| CTR               | 2.28%   |
| Conversion rate   | 32.7%   |
| Cost / conversion | \$15.23 |
| Total conversions | 11,975  |

VS

### ● AdWords

|                    |         |
|--------------------|---------|
| CTR                | 1.28%   |
| Conversion rate    | 10.8%   |
| Cost / conversion  | \$23.97 |
| Avg. views (reach) | 4,771   |

— RECOMMENDATION

## Split budget by objective, not by habit

### VERDICT

Prioritize **Facebook** for conversion-focused campaigns — higher CTR and conversion efficiency lower cost per result. Use **AdWords** for low-cost reach and awareness, where its cheaper clicks and larger footprint matter most. A blended strategy captures each platform's real strength.

— SKILLS DEMONSTRATED

## What this project actually proves

### Data Wrangling

pandas type casting null/dup checks datetime

### Statistical Inference

normaltest Levene's test t-test Mann-Whitney U

### EDA & Visualization

Matplotlib Seaborn correlation matrix  
trend analysis

### Business Translation

metric design cost-per-conversion  
stakeholder framing

## — CONCLUSION

# A decision backed by a p-value, not a preference

This project moved a common marketing decision away from intuition and toward evidence — combining business-metric design with statistically sound hypothesis testing to reach a recommendation leadership can act on with confidence.

Full code & notebook: [github.com/rakeshkpradhan15-hub](https://github.com/rakeshkpradhan15-hub) · [rakeshkpradhan15@gmail.com](mailto:rakeshkpradhan15@gmail.com) · [linkedin.com/in/rakeshkpradhan15](https://linkedin.com/in/rakeshkpradhan15)